For students of art, graphic design and artisan skills such as sculpting and pottery, 3D printing opens up a whole new world of exploration. Projects that would normally take months or years to ideate, conceptualise, draft, mould, redraft, remould and prototype can be done in a much shorter period of time. The physical applications of the project are transferred in to computed aided design programs and are easier to augment, correct and improve before they are made into three dimensional realities.

The most fundamental benefit of 3D printing in the classrooms is that students are taking an active interest in aspects of computer sciences. Since the current use of 3D printers requires at least a basic understanding of CAD programs, students are optimistically taking to freeware like SketchUp to learn how to create simulated versions of their imaginative models. With an increased interest this pathway is leading to a more active engagement with 3D modelling softwares and CAD based programming as well.

3D printers in the classroom open up a whole new world of learning for students.

Medical Use

The use of 3D printing in bio-sciences is still very much in its infancy but theoretical research is still ongoing, to such a degree that scientists have figured out how to print blood vessels using this new technology. But in active practice 3D printing is used in only three categories of health care; prosthetics, medical devices and human tissue.

Prosthetics or "scaffolding" as it is commonly known is already said to have been revolutionized by 3D printing. The ability to cheaply and effectively manufacture join replacements has already been proven. One of the most commonly implemented procedures is knee replacement printing which takes all the advantages of 3D printing. Normally, knee replacement is done from a selection of six base models that doctors have created but with 3D printing, the replacement piece is printed to custom fit the patient who needs it.

By making each replacement knee as unique as the person it belongs to, doctors are able to provide a much better quality of life to their patients. Patients don't lose any bone in the surgery and are able to recover faster and acquire better functionality from the implant. The custom designed bone piece is proven to be stronger, more flexible and a greatly more